Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	**Semester**	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	**Batch**	2023–2027
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Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

Objective

- To implement a **Decision Tree classifier**.
- To extend the Decision Tree into a **Random Forest ensemble model**.
- To study the impact of hyperparameters on overfitting and generalization.
- To **select optimal hyperparameters using 5-Fold Cross-Validation**.
- To compare single-tree and ensemble-tree models.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

Dataset Link: https://archive.ics.uci.edu

Theory

Decision Tree Classifier

A Decision Tree is a supervised learning model that recursively splits the feature space using impurity measures to form decision rules.

Key Concepts:

- Gini Index and Entropy
- Tree depth and node splitting
- Overfitting in deep trees

Limitation: Decision Trees are prone to high variance and overfitting.

Random Forest Classifier

Random Forest is an ensemble learning technique that combines multiple decision trees trained on bootstrapped samples.

Key Ideas:

- Bootstrap aggregation (bagging)
- Random feature selection
- Majority voting

Advantage: Reduces variance and improves generalization compared to a single Decision Tree.

Steps for Implementation

[label=0.]

1. Load the dataset and encode class labels.
2. Perform Exploratory Data Analysis:
   - Class distribution
   - Feature correlation analysis
3. Split the dataset into training and testing sets (80–20).
4. Train a Decision Tree classifier.
5. Define a hyperparameter search space.
6. Perform **5-Fold Cross-Validation** to evaluate each hyperparameter combination.
7. Select the hyperparameters that yield the best average cross-validation performance.
8. Train a Random Forest classifier.
9. Repeat cross-validation-based hyperparameter selection.
10. Compare both models using evaluation metrics.

Hyperparameters to be Explored

Decision Tree

- `criterion`
- `max_depth`
- `min_samples_split`
- `min_samples_leaf`

Random Forest

- `n_estimators`
- `max_depth`
- `max_features`
- `bootstrap`

Hyperparameter Tuning Results

Decision Tree Cross-Fold Results

Table 1: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
gini	5	94.07	0.93

Random Forest Cross-Fold Results

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
50	None	None	96.04	0.94

5-Fold Cross-Validation Performance Comparison

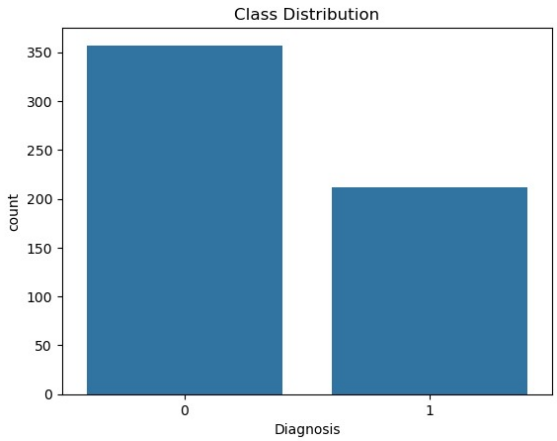
Table 3: 5-Fold Cross-Validation Accuracy Comparison

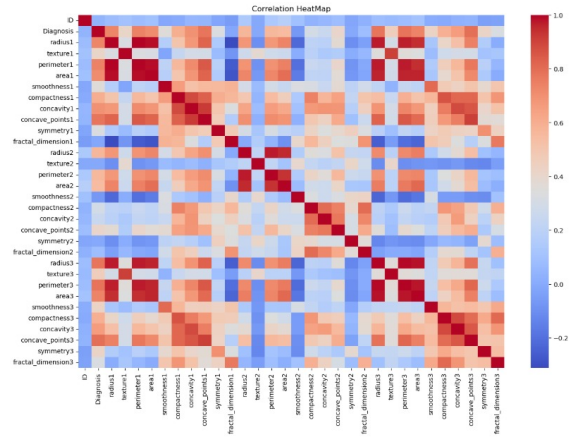
Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9780	0.9451	0.9121	0.9560	0.9121	0.9407
Random Forest	0.9890	0.9560	0.9451	0.9341	0.9780	0.9604

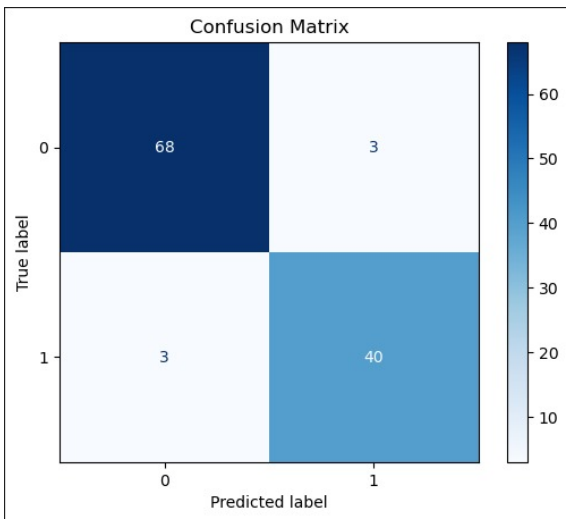
Evaluation Metrics

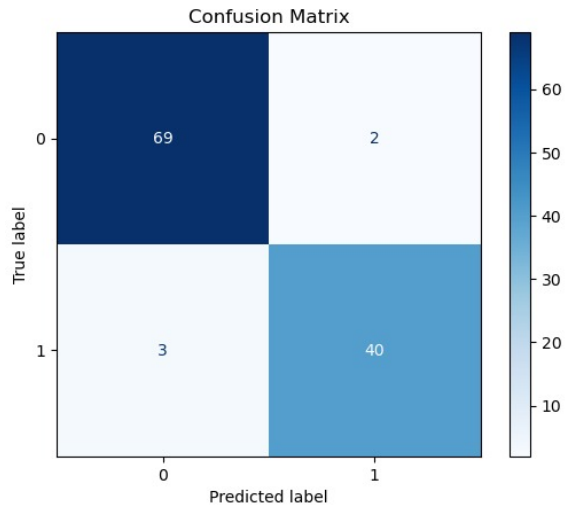
- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve and AUC

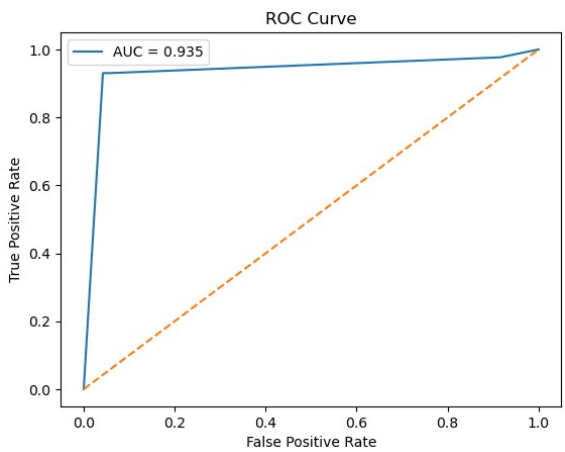
Plots

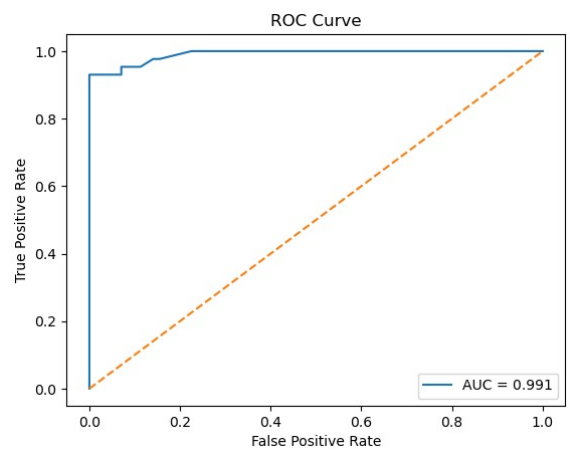

Class Distribution
count
Diagnosis
0
1
350
300
250
200
150
100
50
0


Correlation HeatMap


Confusion Matrix
True label
Predicted label
68
3
3
40
0
1


Confusion Matrix
True label
Predicted label
69
2
3
40
0
1


ROC Curve
AUC = 0.935
True Positive Rate
False Positive Rate


ROC Curve
AUC = 0.991
True Positive Rate
False Positive Rate

Observation Questions

- How does tree depth affect overfitting in Decision Trees?

  Tree Depth plays a significant role in Decision Trees . When the depth is very large , then the model tries to all possible splits which might lead to overfitting instead of generalisation. In the notebook , we have tried (None ,5 ,10 ,20).

- Which hyperparameter had the greatest impact on performance?

  Tree Depth played a major role on the performance ,as it is very much controls the amount of information needed for generalisation. The optimal tree depth gives generalisation while more depth leads to overfitting and while the few numbers leads to underfitting.

- How does Random Forest improve generalization?

  Random Forest is the algorithm of choosing the best suitable Decision Tree with random sampling in order to avoid the overfitting data during the training process.

- Did ensemble learning always improve performance? Why or why not?

  In this notebook, for the given dataset the Random Forest had a small increase in accuracy stating that there is some kind of outliers though this need not be the case for all kinds of dataset , where both the algorithms would give the same accuracy if there are no outliers and in that case we should prefer Decision Tree as the computational complexity of Random Forest is very high when compared to Decision Tree.

Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation. Hyperparameters were selected based on average cross-validation performance, ensuring robust generalization. The results demonstrate that Random Forest reduces variance and improves stability compared to a single Decision Tree.

References

- Scikit-learn: Decision Trees
- Scikit-learn: Random Forest
- UCI Dataset